CHARITE UWATWEMBI

Master of Science in Engineering Artificial Intelligence | Machine Learning and Web Development
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PERSONAL PROFILE STATEMENT

Machine learning and artificial intelligence graduate student at Carnegie Mellon University Africa with a First Class Honours bachelor's degree in Software Engineering and professional experience supporting perception systems at Zipline. Experienced in computer vision data quality, machine learning model development, and data analysis across aviation, autonomy. Built end-to-end AI applications including a portable soil health analyzer and an electronic waste classifier. Seeking AI and machine learning engineering opportunities focused on reliable systems and practical public impact.

EDUCATION

Master of Science in Engineering Artificial Intelligence | **Carnegie Mellon University Africa**, Kigali, Rwanda — **Expected Graduation: May 2028**
Mastercard Foundation Scholar

Bachelor of Science in Software Engineering | **African Leadership University**, Kigali, Rwanda — **Graduated Jul 17, 2026**
Mastercard Foundation Scholar : Machine Learning specialization | First Class Honours | GPA 4.48/5.0

PROFESSIONAL EXPERIENCE

Data Labeling Specialist | **Zipline**, Kigali, Rwanda — **Apr 2025 - Present**
- Label and annotate visual and sensor data used to train and validate perception models for Zipline's autonomous drone navigation and delivery systems.
- Perform quality checks and resolve annotation inconsistencies to maintain accurate and reliable datasets for real-time decision-making.
- Collaborate with perception engineers and cross-functional teams to refine annotation guidelines, address complex edge cases, and support safe and efficient drone operations.

Data Visualization and Mapping Extern | **National Geographic Society and The Nature Conservancy**, Remote (USA) — **Sep 2024 - Nov 2024**
- Analyzed wetland degradation and community health risks using ArcGIS Online, Survey123, and QuickCapture, producing narrative maps and insights for conservation planning.

Academic Intern, Data Analyst | **National Institute of Statistics of Rwanda**, Kigali, Rwanda — **May 2024 - Aug 2024**
- Cleaned, analyzed, and visualized national survey data using Excel and statistical tools, supporting evidence-based reports and research.

Digital Transformation Associate | **United Nations Volunteers and UNDP Rwanda**, Kigali, Rwanda — **Jan 2023 - Dec 2023**
- Supported SharePoint deployment and internal workflow automation for UNDP Rwanda, digitizing office processes and improving access to information.
- Trained 24 staff members on new digital tools, enabling adoption of updated workflows and shared systems.

Digital Ambassadors Program Volunteer | **Rwanda**, Burera District — **Mar 2022 - Dec 2022**
- Delivered digital literacy training to 182 community members, helping participants use Irembo e-services and basic mobile and internet tools.

PROJECTS

SoilSync Portable IoT and ML Soil Health Analyzer | Undergraduate Capstone Project **May 2025 - Aug 2025**

- Built a portable soil health analysis system with a React and TypeScript dashboard, Flask API, Supabase database, SMS access, and virtual sensors for nitrogen, phosphorus, potassium, pH, and moisture.
- Trained a Random Forest classifier on Rwanda soil nutrient data, reaching approximately 82% accuracy and generating fertilizer recommendations from soil and environmental features.
- Designed English and Kinyarwanda interfaces so smallholder farmers can receive recommendations through the web dashboard or a feature phone.

IoT and Machine Learning for Sustainable Waste Management | Independent Research Study **Apr 2023 - Dec 2023**

- Investigated how IoT sensors and machine learning could improve recycling and waste reduction through case studies in the Central African Republic, then produced technical and policy recommendations through a research report.

SKILLS AND CERTIFICATIONS

Technical Skills: Web Development (TypeScript, JavaScript, Python, React, Tailwind CSS, Flask, FastAPI, REST API integrations), Machine Learning (deep learning, computer vision, reinforcement learning, PyTorch, TensorFlow, scikit-learn, pandas, NumPy), Data Analysis (Python, R, SQL, Excel, statistical analysis, data visualization)

Certifications: CMU Africa Bridge Program (Carnegie Mellon University Africa, 2025), CodSoft Student Internship Program (CodSoft, 2024), Network Security (Cisco, 2023), Cybersecurity Essentials (Cisco, 2023).

INTERESTS AND ACHIEVEMENTS

Achievements

- Mastercard Foundation Scholar at Carnegie Mellon University Africa and African Leadership University.
- Placed third in the Mental Health category of iAccelerator Phase 6 as part of the Mindora team, securing USD 7,000 in pre-seed funding.
- Named to the African Leadership University Dean's List for Academic Excellence in 2024.

Interests: Applied AI, computer vision, AI governance, digital agriculture, and technology for public benefit

PUBLICATION

Development of a Portable IoT & ML Soil Health Analyzer,AfricArXiv, 2025 . View Publication